

VC_LIST

Overview

vc_list is a configurable scoreboard component. It allows fast and convenient transactions matching. Built-in procedures allows as global, as in-order, as out-of order matching techniques. When used for building checkers/scoreboard, vc_list component enables significant productivity improvements while reducing amount of code lines and improving code clarity.

Syntax

```
vc_list #(bitw, depth) <list_name> ();
```

Parameters		
bitw		Bit width of data
depth		Maximum depth of list
Access Tasks		
Name	Arguments	Description
reset	No	Resets list, removing all list entries
add	data	- value of data to add Add task adds specified data to the end of the list.
Check_all	data	- data value to be checked against list contents Compares given data value against each one of list entries. In a case of match, removes matching value from the list and asserts "ok" event. Otherwise, asserts "err" event which can be monitored by external error generation process
Check_first	data	- data value to be checked against list contents Compares given data value against the first, the oldest list entry. In a case of match, removes matching value from the list and asserts "ok" event. (FIFO – like operation) Otherwise, asserts "err" event which can be monitored by external error generation process
Check_last	data	- data value to be checked against list contents Compares given data value against the last, the most recent list entry. In a case of match, removes matching value from the list and asserts "ok" event. (FILO-like operation) Otherwise, asserts "err" event which can be monitored by external error generation process
reorder	ptr1, ptr2	- reorder pointers Allows reordering of list entries. Can be used together with "check_last" to match transactions with reordering.
Access Events		
ok		Signals about successful match
err		Signals about match failure

exceeded	"Fires" when anyone of counters exceed predefined maximum limit. Maximum limits can be used as additional test-specific constraints. When it fires, "Exceeded_name" access register provides the name of cover item exceeding maximum limit.	-
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Verilog example

```
`include "vc_list.v"

module test;

integer i;
vc_list #(8,4) list ();

initial begin
    #0;

    // Checking all entries (datapath check)
    // -----
    $display("\nTC info: check_all demonstration");
    for (i=0; i<4; i=i+1)
        #10 list.add(i);

    for (i=0; i<4; i=i+1)
        #10 list.check_all(i);

    // Checking first entry only (FIFO -like operation)
    // -----
    $display("\nTC info: check_first demonstration");
    for (i=0; i<4; i=i+1)
        #10 list.add(i);

    for (i=0; i<4; i=i+1)
        #10 list.check_first(i);

    // Checking last entry only (FILO -like operation)
    // -----
    $display("\nTC info: check_last demonstration");
    for (i=0; i<4; i=i+1)
        #10 list.add(i);

    for (i=3; i>=0; i=i-1)
        #10 list.check_last(i);

    // Checking last entry with reordering
    // -----
    $display("\nTC info: check_last demonstration with reordering");
    for (i=0; i<4; i=i+1)
        #10 list.add(i);

    list.reorder(1,2);

    #10 list.check_last(3);
    #10 list.check_last(1);
    #10 list.check_last(2);
    #10 list.check_last(0);

    #10 $finish;
end

always @(list.err)
    $display("[%0t] <%m> Error: %0d does not match", $time, list.data_reg);

always @(list.ok)
    $display("[%0t] <%m> Info: Match found for %0d", $time, list.data_reg);

endmodule
```